

Matsumura's *Commutative Ring Theory*

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§ 1 Commutative rings and modules

1.1 Ideals

1.1: Let A be a ring and $I \subseteq \text{nil}(A)$ an ideal. If a maps to a unit in A/I then a is a unit.

Proof. Because I is a subset of the nilradical it is contained in every prime ideal. There is a bijection between ideals containing I and ideals in A/I . If $a \mapsto \bar{a} \in A/I$ a unit, then in particular $\bar{a}$ is not contained in any maximal ideal of A/I . Preimages preserve containment, so a is not contained in any ideal containing I of A . However all maximal ideals contain I as they are in particular prime, and so a is not contained in any maximal ideal. Equivalently, a is a unit. $\triangle$

1.2: Let $A_1, \dots, A_n$ be rings; then the prime ideals of $A_1 \times \dots \times A_n$ are of the form $A_1 \times \dots \times A_{i-1} \times P_i \times A_{i+1} \times \dots \times A_n$ for P_i a prime ideal of A_i .

Proof. An ideal of the form $P = A_1 \times \dots \times A_{i-1} \times P_i \times A_{i+1} \times \dots \times A_n$ is clearly prime: If $xy = (x_1y_1, \dots, x_ny_n) \in P$, then we have $x_j, y_j \in A_j$ for all $j \neq i$, and at least one of x_i, y_i in P_i by primality. Therefore one of $(x_1, \dots, x_n)$ or $(y_1, \dots, y_n)$ is in P . To see that all primes are of this form, let P be a prime ideal in the product. Let $x = (x_1, \dots, x_n)$ be in P . We can factor this element in many ways: In fact, we may write the element x in the form $(x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_n)(1, \dots, x_i, \dots, 1)$ for any i . Primality says that one of these factors must lie in P , and it cannot be the case that the former factor lies in P for all i , or else $P = A_1 \times \dots \times A_n$. Therefore there must be some i such that $(1, \dots, x_i, \dots, 1) \in P$. In fact there can be only one such i : If $(1, \dots, x_j, \dots, 1) \in P$ for some $i \neq j$ then the unit in the i th coordinate generates all of A_i , contradicting the fact that P is proper. Then the fact that $P = A_1 \times \dots \times A_{i-1} \times P_i \times A_{i+1} \times \dots \times A_n$ follows by primality of P . $\triangle$

1.3: Let A and B be rings, and $f : A \rightarrow B$ a surjective homomorphism. (a) Prove that $f(\text{rad } A) \subseteq \text{rad } B$, and construct an example where the inclusion is strict. (b) Prove that if A is semilocal then $f(\text{rad } A) = \text{rad } B$.

Proof. Part (a): Recall that $\text{rad } B = \bigcap_{\mathfrak{m} \text{ maximal in } B} \mathfrak{m}$. Now note that the preimage of a maximal ideal under a surjective map is also maximal, so $f^{-1}(\text{rad } B) = f^{-1}(\bigcap \mathfrak{m}) = \bigcap f^{-1}(\mathfrak{m}) = \bigcap_{\ker f \subseteq \mathfrak{n} \text{ maximal in } A} \mathfrak{n} \supseteq \text{rad } A$. That is, $\text{rad } A \subseteq \{x \in A \mid f(x) \in \text{rad } B\}$, so $f(\text{rad } A) \subseteq \text{rad } B$.

Part (b): We induct on the number of maximal ideals in A , which we denote m . If $m = 1$ then A is local and $\text{rad } A = \mathfrak{m}$. We have that $\ker f$ is an ideal, so in particular it has a maximal ideal containing it which must be $\mathfrak{m}$. Now our bijection between maximal ideals containing $\ker f$ and maximal ideals of B shows that B is local and so $\text{rad } A = \mathfrak{m} = f^{-1}(\text{rad } B)$. Now $\text{rad } A = \{x \mid f(x) \in \text{rad } B\}$ and $f(\text{rad } A) = \text{rad } B$. Now suppose that A has m maximal ideals and for any $m - 1$ of these we have $f(\bigcap_{i=1}^{m-1} \mathfrak{m}_i) = \bigcap_{i=1}^{m-1} \mathfrak{n}_i$ where $f^{-1}(\mathfrak{n}_i) = \mathfrak{m}_i$. Now $f^{-1}(\text{rad } B) = f^{-1}(\mathfrak{n}_m \cap \bigcap_{i=1}^{m-1} \mathfrak{n}_i) = \mathfrak{m}_m \cap \bigcap_{i=1}^{m-1} \mathfrak{m}_i = \text{rad } A$ and so the result follows. $\triangle$

1.4: Let A be an integral domain. Then A is a UFD if and only if every irreducible element is prime and the principal ideals of A satisfy the ascending chain condition. (Equivalently, every non-empty family of principal ideals has a maximal element).

Proof. ($\Rightarrow$) Suppose A is a UFD. Then any non-zero, non-unit element $a \in A$ has a unique expression as a product of prime elements. If a is irreducible, it is given by a product of one prime, or else irreducibility is violated. Now consider a family of principal ideals $\{(a_i)\}_{i \in I}$ in A with $a_i \subseteq a_{i+1}$, and note that we may write each a_i uniquely as $a_i = \prod_j p_{i,j}$ for $p_{i,j}$ primes. The fact that $a_i \subseteq a_{i+1}$ implies that $a_i \in (a_{i+1})$, so $a_i = ba_{i+1}$. Then a_i 's prime factorization must be a product of a_{i+1} 's prime factorization with b 's prime factorization. We may continue this procedure, strictly decreasing the number of prime elements in the expression of a_j . This process must halt, as we will eventually eliminate all primes in the factorization except one, p , which will persist as it is irreducible.

($\Leftarrow$) Suppose every irreducible element of A generates a prime ideal and that every family of principal ideals satisfies the ascending chain condition. If $a \in A$ is irreducible it has an expression as a product of primes: a . Suppose a is reducible, so $a = bc$ for some $b, c \in A$. We continue this process until we have expressed a as a product of irreducibles. This must be a finite product, as principal ideals satisfy the ascending chain condition. This gives an expression for a as a product of primes. $\triangle$

1.5: Let $\{P_\lambda\}_{\lambda \in \Lambda}$ be a non-empty family of prime ideals, and suppose that the P_λ are totally ordered with respect to inclusion; then $\bigcap P_\lambda$ is a prime ideal. Also, if I is any proper ideal, the set of prime ideals containing I has a minimal element.

Proof. Note that the intersection is nonempty by the fact that $\{P_\lambda\}_{\lambda \in \Lambda}$ is totally ordered. Suppose $a, b \notin \bigcap P_\lambda$. Then in particular $a, b \notin P_\lambda$ for any $\lambda \in \Lambda$. However now $ab \notin P_\lambda$ for any $\lambda \in \Lambda$, which implies that $ab \notin \bigcap P_\lambda$. Therefore $\bigcap P_\lambda$ is prime. Now let I be any proper ideal, and consider the set of all prime ideals containing I . There is at least one maximal ideal containing I , so the intersection $J = \bigcap_{I \subseteq P} P$ is nonempty. Then the same argument above applies: If $a, b \notin J$ then in particular $a, b \notin P$ for any $P \supseteq I$. Then $ab \notin P$ for any P by primality, and so J is prime and minimal with respect to inclusion by construction. $\triangle$

1.6: Let A be a ring, $I, P_1, \dots, P_r$ ideals of A , and suppose that $P_1, \dots, P_r$ are prime, and that I is not contained in any of the P_i ; then there exists an element $x \in I$ not contained in any P_i (solved with hints from ChatGPT).

Proof. For $r = 1$ we choose any $x \in I - P_1$ and we are done.

For $r = 2$ let $x_i \in I - P_i$, and let $x = x_1 + x_2$. If neither element is in P_1 , nor are they in P_2 , then we are finished. If one element is in one ideal, say $x_1 \in P_2$, then we have that the image of x in A/P_2 is x_2 , which is nonzero, $x_1 + x_2 \notin P_2$. Likewise, if $x_2 \in P_1$, the same argument shows that the sum is still nonzero. It cannot be the case that both are contained in the same prime ideal. Therefore the $r = 2$ case holds.

Now we induct on the number of prime ideals. The above argument shows that the statement holds for zero prime ideals, so suppose it holds for $(r - 3)$ -many. That is, there is some $x \in I$ such that $x \notin P_i$ for $1 \leq i \leq r - 1$. Note that we may assume $P_i \not\subseteq P_j$ for all $1 \leq i < j \leq r$, as if $P_i \subseteq P_j$ then x avoiding P_j implies that it avoids P_i . If $x \notin P_r$ we are done. If $x \in P_r$ consider $x_i \in P_i - P_r$ for each i , which exist by our condition that no P_i contains any other. Then for $y \in I$ we see that $x + y \prod x_i \in I$ and is nonzero in the quotient ring A/P_i for all $1 \leq i \leq r - 1$ by the fact that $x \notin P_i$. This sum being zero in A/P_r implies that $x = -\prod x_i$, which in turn implies that some $x_i \in P_r$ by primality. However this cannot be the case by supposition, so the inductive step holds and the claim is proven. $\triangle$

1.2 Modules